

Effective and Fast: A Novel Sequential Single Path Search for Mixed-Precision Quantization

Qigong Sun, Licheng Jiao, Yan Ren, Xiufang Li, Fanhua Shang, Fang Liu

Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Joint International Research Laboratory of Intelligent Perception and Computation, School of Artificial Intelligence, [Xidian University](#), Xi'an, Shaanxi Province 710071, China

xd.qigongsun@163.com, lchjiao@mail.xidian.edu.cn, yanren@stu.xidian.edu.cn, xfl_xidian@163.com, fhshang@xidian.edu.cn, f63liu@163.com

Abstract

Since model quantization helps to reduce the model size and computation latency, it has been successfully applied in many applications of mobile phones, embedded devices and smart chips. The mixed-precision quantization model can match different quantization bit-precisions according to the sensitivity of different layers to achieve great performance. However, it is a difficult problem to quickly determine the quantization bit-precision of each layer in deep neural networks according to some constraints (e.g., hardware resources, energy consumption, model size and computation latency). To address this issue, we propose a novel sequential single path search (SSPS) method for mixed-precision quantization, in which the given constraints are introduced into its loss function to guide searching process. A single path search cell is used to combine a fully differentiable supernet, which can be optimized by gradient-based algorithms. Moreover, we sequentially determine the candidate precisions according to the selection certainties to exponentially reduce the search space and speed up the convergence of searching process. Experiments show that our method can efficiently search the mixed-precision models for different architectures (e.g., ResNet-20, 18, 34, 50 and MobileNet-V2) and datasets (e.g., CIFAR-10, ImageNet and COCO) under given constraints, and our experimental results verify that SSPS significantly outperforms their uniform counterparts.

1. Introduction

With the development of deep neural networks (DNNs), DNNs have achieved impressive performance in many applications. Generous computing resources and footprint memory are urgently required when deeper networks are used to solve various problems. Moreover, with the rapid development of chip technologies, especially GPU and

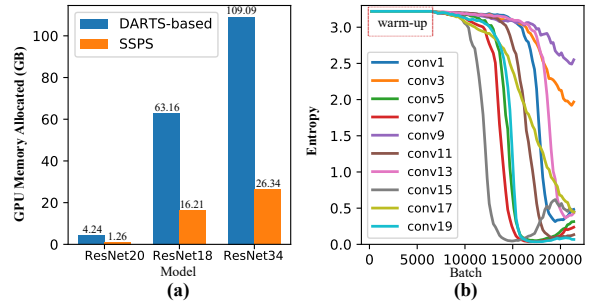


Figure 1: (a) Comparison of the GPU memory usage of searching process on multiple models based on the DARTS-based method and SSPS, where each search cell has 5 candidates (i.e., 2-bit to 6-bit) and batch size is 32. (b) The searching entropy variation curves of some layers in ResNet-20 on CIFAR-10.

TPU, the computational frequency and efficiency have been greatly improved. Most scholars use GPUs as the basic hardware platform for network training due to excellent acceleration capability. However, for low power consumption platforms (e.g., mobile phone, embedded devices and smart chips), whose resources are limited, it is hard to achieve satisfactory performance for industrial applications. As one of the typical methods for model compression and acceleration, model quantization methods usually quantize full-precision (32-bit) parameters to low-bits of precision (e.g., 8-bit and 4-bit). Even more, we can extremely constrain weights and activation values to binary $\{-1, +1\}$ [18, 40] or ternary $\{-1, 0, +1\}$ [61], which can be computed by bitwise operations. This logic calculation is more suitable for the implementation of FPGA and other service-oriented computing platforms. As implemented in [27], they achieved about $705\times$ speed up compared with CPU and are $70\times$ faster than GPU in the peak condition.

In recent years, many methods [8, 22, 26, 30, 46, 58] have been proposed to improve the performance of low-precision models. However, the bit-precisions of most quan-